Understanding the 'Unknown': A Vision-Language Framework for Context-aware Safety Monitoring in Mobile Robots

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ABSTRACT

Autonomous mobile robots deployed in construction environments must perform safety-critical assessments under frequent occlusion, viewpoint variation, and distance-induced perceptual degradation. Under such conditions, visual evidence is often incomplete or ambiguous, yet conventional perception systems tend to produce overconfident decisions. This work addresses the problem of safety assessment under partial observability by introducing an explicit unknown state that represents insufficient visual evidence rather than low prediction confidence. We formulate safety evaluation as a relational decision problem and propose a structured vision-language reasoning framework that separates visual cue extraction from decision regulation. The framework employs structured prompting to extract intermediate visual cues and evaluates relational consistency to determine whether available evidence is sufficient to support a definitive safety judgment. When evidence is inadequate, the system defers the decision by outputting an unknown state. Experiments conducted in a controlled simulation environment for lifeline connectivity assessment demonstrate that the proposed approach reduces overconfident safety decisions compared to direct prompting. Furthermore, analysis of human annotation entropy reveals spatially structured perceptual difficulty patterns, and the model's uncertainty responses align with human uncertainty patterns. These results suggest that modeling perceptual sufficiency enables more interpretable and self-aware safety monitoring behavior in mobile robotic systems.

INTRODUCTION

The introduction of mobile robots into construction sites has created new opportunities for autonomous inspection and safety monitoring in highly dynamic environments (Liang et al. 2021). To adapt to continuously changing surroundings, mobile robots rely heavily on visual perception for navigation, obstacle avoidance, and autonomous decision-making in dynamic environments. Recent advances in robot vision have enabled autonomous systems to detect safety-related elements and interpret environmental conditions through visual sensing (Prieto et al. 2020; Robinson et al. 2023). However, construction environments impose substantial visual challenges, including frequent occlusion, significant viewpoint variation, and large distance changes (Li and Li 2022). Under such conditions, available visual evidence may be incomplete or ambiguous.

However, current vision-based perception systems, largely built upon object-centric detection and classification frameworks that infer safety conditions from independently detected elements, often produce overconfident safety judgments, even when the visual input does not sufficiently support reliable decision-making (Wang et al. 2024).

Conventional vision-based approaches attempt to mitigate visual ambiguity through data augmentation and robust training strategies (Chen et al. 2023; Li and Li 2022). However, their performance remains limited when safety-critical relationships are only partially observable. To address these limitations, active vision strategies have been proposed in robotics, enabling robots to adjust their viewpoints to improve scene observability (Tosidis et al. 2022). In prior work, an active vision framework for construction monitoring was proposed to improve scene observability through viewpoint adjustment, including angle-aware path planning and field-of-view–guided exploration (Park et al. 2024; Yang et al. 2025). Nevertheless, existing active vision frameworks remain predominantly evidence-driven and do not explicitly model perceptual sufficiency. Consequently, they lack a principled means to evaluate the adequacy and reliability of the available visual evidence. This limitation hinders the robot's ability to establish a coherent assessment of the current perceptual state, potentially resulting in unreliable safety judgments or unnecessary viewpoint adjustments in safety-critical contexts.

To address this limitation, a structured *unknown* state is introduced to enable the robot to defer safety-critical judgments when visual evidence is insufficient, thereby supporting principled decision-making under partial observability. The aim of this study is to equip mobile robots with the ability to recognize insufficient or ambiguous evidence and to regulate decisions in an interpretable manner. This objective is achieved through a VLM-based structured visual reasoning framework that explicitly models relational safety conditions and incorporates the proposed unknown state. Experimental evaluations demonstrate that the proposed approach mitigates overconfident safety decisions and produces structured uncertainty responses. Furthermore, the resulting uncertainty patterns are analyzed to examine their alignment with human perceptual uncertainty, providing insight into the interpretability and plausibility of the model's behavior.

BACKGROUND

Perception for Autonomous Monitoring Robots in Construction. Autonomous monitoring robots in construction primarily rely on vision-based perception to identify safety-relevant elements and assess regulatory compliance (Kim et al. 2022; Zhang and Zhang 2022). Existing approaches largely adopt object-centric detection frameworks, in which individual safety items such as helmets, harnesses, or hazardous worker postures are detected independently using supervised learning models (Zhou et al. 2021). While such methods have demonstrated promising performance under controlled conditions, their effectiveness deteriorates in complex construction environments characterized by occlusion, viewpoint variation, and scale changes. In safety-critical monitoring, however, assessment frequently requires relational reasoning beyond isolated object detection. For example, verifying fall protection compliance depends not only on detecting a lifeline, but also on confirming its valid connection between a harness and an anchor structure. Conventional detection pipelines are not explicitly designed to reason about such structured relationships, particularly when observations are partial or occluded (Battaglia et al. 2018; Su et al. 2025). Without structured representations of perceptual sufficiency, missing observations may be misinterpreted as either confirmed absence or incomplete evidence requiring further exploration. Such ambiguity can undermine reliable safety assessment and lead to inappropriate decisions or unnecessary viewpoint adjustments.

Uncertainty Representation in Vision-Based Perception. Uncertainty-aware perception has been widely studied to improve prediction reliability under ambiguous or noisy observations (Deng et al. 2022). In vision-based systems, uncertainty is commonly represented through prediction confidence or probabilistic estimates derived from model outputs (Guo et al. 2017). However, scalar confidence values do not necessarily indicate whether the available visual evidence is sufficient for reliable reasoning. Under partial observability, incomplete observations often produce low detection confidence that may be incorrectly interpreted as object absence, making scalar confidence insufficient for distinguishing between genuine absence and insufficient observation. To address this limitation, prior work has introduced abstention or reject-option mechanisms that withhold predictions under low confidence (Gawlikowski et al. 2023). Although effective for reducing prediction errors, these methods remain grounded in confidence-based uncertainty and do not address perceptual sufficiency in structured reasoning (Geifman and El-Yaniv 2019).

Vision-Language Models for Structured Visual Reasoning. Vision-language models (VLMs) enable joint visual and linguistic interpretation beyond object-level detection, allowing relational and contextual descriptions of complex scenes. However, existing vision-language applications primarily aim to generate scene interpretations or definitive task decisions from observed inputs (Alayrac et al. 2022; Radford et al. 2021), and have been shown to struggle with compositional and relational reasoning under incomplete or ambiguous information (Cascante-Bonilla et al. 2024), without explicitly modeling perceptual sufficiency. In safety-critical monitoring, distinguishing confirmed absence from insufficient observation can help reduce the risk of premature or overconfident decisions. Therefore, while VLMs provide a foundation for relational scene interpretation, they do not inherently regulate decisions based on evidential adequacy. This motivates a framework that separates scene interpretation from decision regulation and enables explicit judgment deferral under incomplete observation.

METHODOLOGY

Problem Formulation. This study formulates safety assessment in robotic monitoring as a relational decision problem under partial observability. Given a visual observation I captured from a dynamic viewpoint, the robot must determine a safety state $y \in \{\text{connected, not connected, unknown}\}$. The decision concerns whether a connectivity relationship holds between safety components, rather than merely whether individual objects are detected. The unknown state represents insufficient visual evidence to establish this relational condition. Unlike conventional perception pipelines that interpret missing detections as object absence, the proposed formulation distinguishes between true disconnection and incomplete observation of the connectivity relationship. Accordingly, safety evaluation requires both relational interpretation and an explicit assessment of whether the connectivity relationship can be reliably established. To operationalize this formulation, we adopt the Lifeline Connectivity Task Assessment, which evaluates whether a safety harness is securely connected to a valid anchor point. When the connectivity relationship cannot be established from the current observation, the system outputs the explicit unknown state.

Structured Visual Reasoning Framework. The proposed framework adopts a structured visual reasoning architecture that integrates vision-language modeling with structured relational decision regulation. Rather than relying on end-to-end classification, the framework decomposes safety assessment into intermediate visual cue extraction followed by relational consistency evaluation.

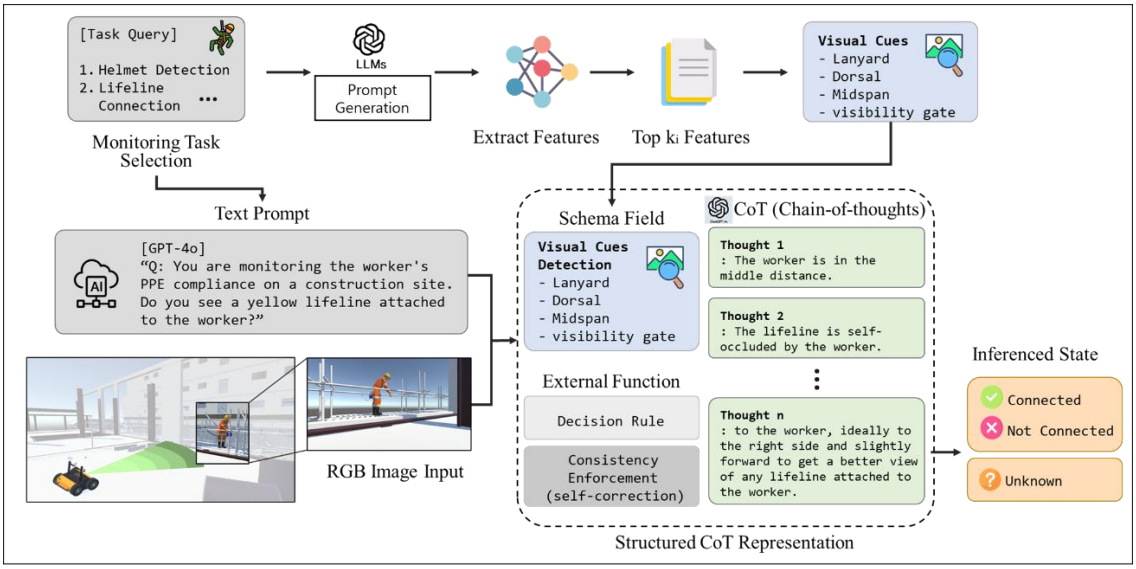


Figure 1. Structured Visual Reasoning Framework.

Given an input RGB image, a vision–language model (VLM) is prompted with a structured query to extract safety-relevant perceptual cues instead of directly predicting a safety state. The model identifies semantically meaningful visual components required for safety monitoring, including equipment presence, spatial continuity, visibility conditions, and occlusion characteristics, and outputs them in a predefined JSON representation. This intermediate cue representation decomposes safety assessment into interpretable perceptual elements, avoiding overconfident end-to-end classification while allowing cue definitions to adapt to different safety objectives.

Prior to reasoning, extracted cues undergo a normalization step in which raw VLM outputs are mapped into predefined discrete states for each perceptual cue, ensuring semantic consistency across extracted descriptions. In particular, occlusion attributes are normalized using visibility assessments and midspan region clarity, preventing contradictory perceptual interpretations from propagating into downstream reasoning. The normalized cues are processed by a structured relational reasoning module that evaluates relationships among cues rather than treating them independently. Within this framework, safety decisions emerge from the relational consistency of perceptual cues, where logical interactions among observed conditions determine the outcome instead of manually specified decision rules or individual cue predictions.

A key feature of the framework is explicit modeling of perceptual sufficiency. The system evaluates not only relational compliance but also whether available visual evidence is sufficient to support a reliable safety-critical judgment. The unknown state is produced when perceptual evidence is incomplete or insufficient to establish relational continuity, enabling controlled decision deferral under partial observability and reducing premature safety conclusions. To further ensure safety reliability, a post-decision consistency validation stage is applied after relational reasoning. This stage verifies whether relational continuity and supporting evidence remain consistent across extracted cues. This additional verification step prevents overconfident positive judgments and reinforces safety-aware reasoning under uncertain perceptual conditions.

EXPERIMENTS

Dataset Collection. To enable controlled evaluation under systematically varying observability conditions, a 3D Unity-based simulation environment was developed instead of real-world data

collection. This environment allows precise manipulation of viewpoint, distance, and occlusion factors that are difficult to regulate consistently in construction sites. The simulator includes worker models equipped with safety harnesses and lifelines within scaffold structures, where perceptual difficulty primarily arises from pose-dependent self-occlusion caused by variations in worker orientation. Difficulty was varied through robot–worker distance and viewing angle (left, front, right). The dataset targets lifeline connectivity assessment, determining whether a safety harness is attached to a valid anchor point. Connectivity states are defined as left (L), Rear (M), right (R) attachment, or not connected (N), representing compliant and non-compliant safety conditions. Approximately 10,000 RGB images were generated by varying viewpoint, distance, and pose-induced self-occlusion, producing monitoring scenarios under partial observability.

Human Annotation Protocol. To establish ground truth for safety assessment, three human annotators independently labeled each image in the dataset as connected, not connected, or unknown. The unknown label was assigned when visual evidence was insufficient to confidently determine connectivity. Annotation guidelines explicitly distinguished between true absence and insufficient observation due to occlusion. Images were presented in randomized order, and annotations were collected without discussion to preserve natural disagreement.

Experimental Setup. The proposed framework was implemented using GPT-4o as the underlying vision-language model. Structured prompts were designed to guide intermediate visual cue extraction, and the extracted outputs were parsed into a predefined JSON schema for relational decision evaluation. The evaluation examines whether the reasoning framework aligns its predictions with human uncertainty patterns under partial observability by assessing whether the model outputs the unknown state when human annotators exhibit uncertainty. As a baseline model for comparison, a direct prompting approach was implemented in which GPT-4o directly predicts task labels (connected / not connected / unknown) without structured relational reasoning.

Evaluation Metrics. Human perceptual uncertainty is quantified using Shannon entropy over annotator label distributions. For each image x, uncertainty is computed as:

$$H_{human}(x) = -\sum_{y} p_{human}(y \mid x) \cdot \log p_{human}(y \mid x),$$

where $p_{human}(y \mid x)$ denotes the empirical label distribution. Higher entropy indicates greater human disagreement and higher perceptual difficulty. Entropy values are mapped onto the viewpoint lattice to construct a spatial human uncertainty map for evaluating model self-awareness. Soft Accuracy measures prediction correctness while assigning greater weight to low-entropy (clear) samples and allowing unknown predictions on high-entropy (ambiguous) samples, where soft labels are defined by the empirical annotation distributions obtained from human annotators. Unknown Rate measures the proportion of predictions assigned to the unknown state, reflecting the model's ability to avoid overconfident decisions under insufficient visual evidence.

Results. As illustrated in Figure 2(a), annotation entropy exhibits a structured spatial distribution rather than a random pattern. Uncertainty increases when the upper-central region containing the observed worker becomes self-occluded or visually degraded. Across connectivity cases, high-entropy regions align with viewpoints that occlude the connectivity area. Consensus maps further show consistent agreement on the unknown label under limited relational visibility, indicating systematic human judgment deferral.

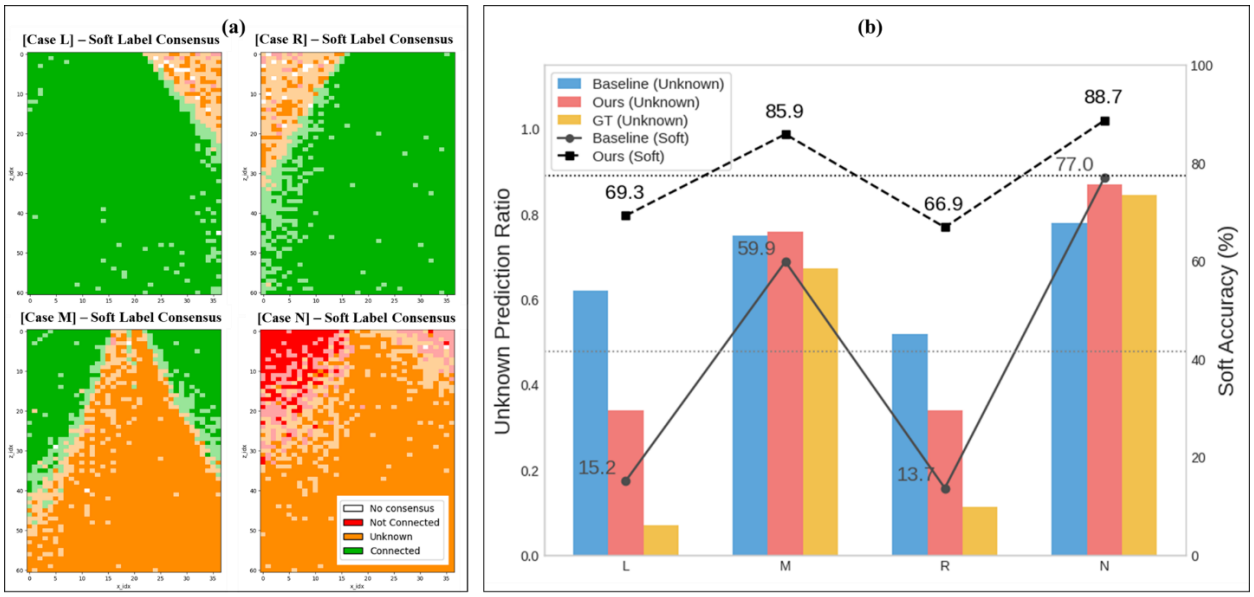


Figure 2. (a) Spatial Map of Annotation – Spatial distribution of human annotation uncertainty over the viewpoint–distance lattice. Each cell represents a robot viewpoint (horizontal: viewing angle; vertical: robot–target distance), where color intensity indicates entropy derived from human annotations and consensus maps show the dominant label. **(b) Unknown Ratio and Soft Accuracy by Cases –** Comparison between the baseline and the proposed framework in terms of unknown prediction ratio (bars) and soft accuracy (lines) across cases: L (Left attachment), M (Rear attachment), R (Right attachment), and N (Not connected).

Figure 2(b) compares prediction behaviors between the baseline direct prompting approach and the proposed structured reasoning framework, revealing distinct differences in the regulation of unknown predictions across connectivity cases. Rather than uniformly increasing abstention, the proposed framework selectively regulates the use of the unknown state across connectivity cases. Unknown predictions decrease in clearer configurations (Case L: 0.62 to 0.34, Case R: 0.52 to 0.34) while increasing or remaining comparable in perceptually ambiguous cases (Case M: 0.75 to 0.76, Case N: 0.78 to 0.87), closely matching the ground-truth unknown distribution. In contrast, the baseline maintains relatively high unknown rates even in clearer views while issuing definitive predictions under ambiguity, indicating inconsistent decision regulation without explicit modeling of perceptual sufficiency. This redistribution coincides with substantial improvements in soft accuracy across all cases, increasing from 15.2 to 69.3 (L), 59.9 to 85.9 (M), 13.7 to 66.9 (R), and 77.0 to 88.7 (N), yielding an overall improvement from 41.5 to 77.5. These gains indicate that decisions are deferred specifically when relational continuity cannot be reliably established, rather than through increased conservatism alone.

As robot–target distance increases, the unknown prediction ratio also rises, indicating that the system increasingly withholds judgments as visual observability degrades. Unknown predictions spatially correspond to regions of elevated human annotation entropy, suggesting that decision deferral occurs under similar perceptual constraints. While the model exhibits a slightly more conservative tendency than human annotators, it reduces unnecessary abstention in regions with clear visibility and produces definitive assessments when sufficient evidence is available. Despite these improvements, several failure modes remain. Errors primarily occur under severe occlusion angles where critical relational cues become visually indistinguishable even for human observers. In such cases, the vision–language model occasionally produces incomplete cue

extraction, leading to incorrect relational evaluation. However, compared with the baseline, these failures more often result in unknown predictions rather than incorrect definitive judgments.

DISCUSSION AND CONCLUSION

The observed behavioral patterns indicate that the proposed framework regulates safety decisions by separating perceptual interpretation from decision commitment under partial observability. Unlike conventional perception pipelines that directly produce categorical predictions or rely on confidence thresholds, the structured reasoning process evaluates relational evidence through intermediate visual cues before issuing a final decision. Consequently, degraded or incomplete observations lead to controlled decision deferral rather than premature categorical judgments, producing uncertainty behavior that more closely reflects human perceptual uncertainty.

This study introduced a structured visual reasoning framework with an explicit unknown state that models perceptual sufficiency as a distinct decision criterion. Experimental results demonstrate that the framework not only improves categorical prediction performance but also adapts its decisions according to perceptual difficulty, generating unknown predictions in genuinely ambiguous regions while reducing premature definitive judgments observed in direct prompting approaches. From a safety monitoring perspective, uncertainty therefore functions as an evidence-based decision state rather than a byproduct of prediction confidence, enabling more interpretable and safety-aware monitoring behavior. Beyond the lifeline connectivity task, the framework is extensible to broader safety-critical perception problems requiring relational reasoning under partial observability. Because perceptual cue extraction is separated from relational decision regulation, new safety objectives can be incorporated by redefining relational cues without retraining task-specific classifiers. The explicit unknown state further provides a natural interface to active vision, where decision deferral signals insufficient evidence and can trigger viewpoint adjustment to acquire more informative observations, linking perception with motion through evidential sufficiency. Nevertheless, this study is conducted in a simulated environment and focuses on a single relational safety task. Real-world construction environments introduce additional variability, including sensor noise, dynamic occlusion, and complex scene interactions. Future work will extend the framework to real-world datasets and broader relational safety scenarios while exploring tighter integration with active perception for adaptive viewpoint planning. In summary, structured vision–language reasoning with explicit unknown-state modeling provides a principled approach for handling ambiguity in safety-critical robotic monitoring. By treating uncertainty as an assessment of evidential sufficiency rather than prediction confidence alone, the framework enables controlled decision deferral and aligns machine decision behavior with human perceptual uncertainty under partial observability.

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